

Miko Decomposition, Hash Chain Commitment, and Mira RLC

Three Deterministic Primitives for Post-Quantum Polynomial Proofs over Binary Fields

Miko Angelo De La Paz
delapazmikoangelo@gmail.com

June 2026

Abstract

We present three deterministic primitives for transparent, post-quantum zero-knowledge proofs over $\mathbb{F}_{2^{256}}$. **Miko Decomposition** uniquely factorises any polynomial of degree $< n = 2^k$ into k quotient polynomials with exponentially decreasing degrees; its correctness is checked via the Miko identity using only two challenge points, achieving $(n/2^{256})^2$ soundness. **Hash Chain Commitment** replaces Merkle trees with sequential SHA-256 hashing while deliberately leaving the residual polynomial un-committed; binding comes algebraically from the Miko identity and Mira cross-binding, and a Phase 3 commitment lock enables native recursion with no adaptive-attack window. **Mira** (Multi-Instance Recursive Aggregation) uses random linear combinations to compress evaluations, cross-bind public and challenge points, and verify recursive Miko towers in $O(1)$ time per tower. The resulting dual-binding architecture (algebraic via Schwartz-Zippel, cryptographic via SHA-256) ensures that even if SHA-256 is broken, algebraic security survives. Together, the primitives yield PMM, a post-quantum ZK proof system with ~ 140 -byte constant-size on-chain proofs, no trusted setup, and native recursive composition.

1 Introduction

Zero-knowledge proof systems require three fundamental operations: verifying that a committed polynomial has bounded degree, binding the prover to that polynomial before challenges are revealed, and compressing verification checks to minimize proof size. Existing approaches use probabilistic queries with tree-based commitments, or algebraic constructions over prime fields requiring trusted setup.

This paper introduces three primitives that perform these operations deterministically over binary fields, without trusted setup, and with hash-independent algebraic security.

1.1 Our Contribution

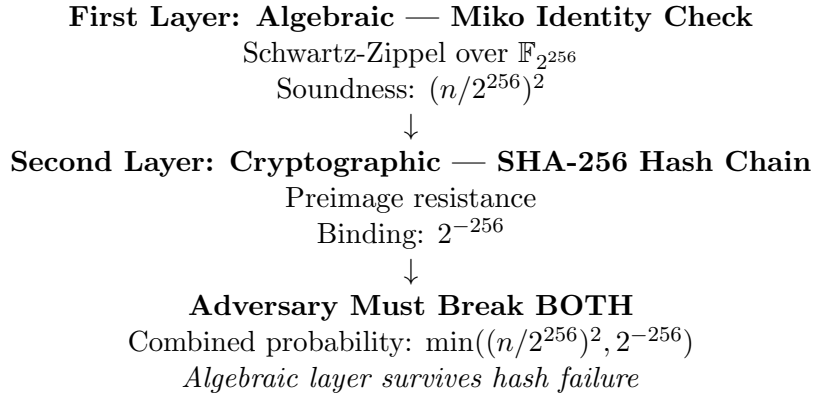
1. **Miko Decomposition (Section 3):** Deterministic polynomial factorization with exponentially decreasing degree bounds. Replaces probabilistic degree testing (FRI) with algebraic verification using the Miko identity. Requires 2 challenge points versus FRI's 30-50. Soundness: $(n/2^{256})^2$.

2. **Hash Chain Commitment (Section 4):** Sequential SHA-256 binding where the residual polynomial is not committed — binding is algebraic via Miko identity and Mira cross-binding. The Phase 3 commitment lock (Lemma 4.3) prevents adaptive attacks in recursive composition. Provides hash-independent security: if SHA-256 breaks, algebraic binding survives.
3. **Mira (Multi-Instance Recursive Aggregation) (Section 5):** RLC meta-protocol for evaluation compression and cross-binding. Compresses s challenge evaluations per quotient into one field element. Compresses m RMC internal node checks into $O(1)$ verification. Enables constant-size proof aggregation across multiple proofs.

1.2 First Application: PMM

These primitives compose into PMM (Polynomial Miko-Mira), a transparent post-quantum ZK proof system demonstrating native recursion (Section 8). PMM serves as the reference application, not the sole use case. The primitives are general — any ZK system over characteristic 2 can adopt them independently.

1.3 The Dual Binding Security Architecture



1.4 Comparison to Prior Approaches

2 Preliminaries

2.1 The Binary Field $\mathbb{F}_{2^{256}}$

Elements are 256-bit polynomials over $\mathbb{F}_2$ modulo the irreducible pentanomial:

$$p(X) = X^{256} + X^{10} + X^5 + X^2 + 1$$

Addition is XOR (single cycle). Multiplication is carryless with Barrett reduction, hardware-accelerated via PCLMULQDQ (x86-64) and PMULL (ARM64). Squaring is the Frobenius endomorphism: $(a+b)^2 = a^2 + b^2$ (free bit-spreading). Inversion uses Itoh-Tsujii with ~ 15 -step addition chain.

No discrete logarithm: The additive group has exponent 2 ($a + a = 0$ for all a). The multiplicative group $\mathbb{F}_{2^{256}}^\times$ is cyclic of order $2^{256} - 1$ but is never used for cryptographic hardness.

Table 1: Comparison of Proof Systems

Property	FRI (STARKs)	Merkle Trees	Miko Decomposition	Hash Chain + RMC
Degree testing	Probabilistic, 30–50 queries	N/A	Deterministic, 2 challenges	N/A
Commitment structure	Requires Merkle trees	Generic hash tree	N/A	Sequential chain + recursive Miko
Algebraic meaning	None (proximity only)	None (opaque hashes)	Miko identity at every level	Miko identity at every RMC node
Hash-independent?	No	No	Yes	Yes (dual binding)
Selective opening	Required (query positions)	Core purpose	Not needed	Not needed
Verification cost	$O(\text{queries} \cdot \log n)$	$O(\log n)$ per path	$O(n)$ total	$O(1)$ per tower
Proof size	50–200 KB	$O(\log n)$ per path	~ 400 B (inner)	~ 140 B (outer)

2.2 The Schwartz-Zippel Lemma

Lemma 2.1 (Schwartz-Zippel). *For any non-zero polynomial $f(X)$ of degree d over field $\mathbb{F}$ and uniformly random $z \in \mathbb{F}$:*

$$\Pr[f(z) = 0] \leq \frac{d}{|\mathbb{F}|}$$

This bound is information-theoretic — no quantum algorithm can increase it.

For s independent challenges chosen after the polynomial is fixed via Fiat-Shamir commitment:

$$\Pr[\forall i : f(z_i) = 0 \mid f \neq 0] \leq \left(\frac{d}{|\mathbb{F}|} \right)^s$$

Table 2: Parameter-Dependent Soundness Error for $s = 2$

Circuit Size n	Soundness $(n/2^{256})^2$
$2^{10} = 1,024$	2^{-492}
$2^{13} = 8,192$	2^{-486}
$2^{15} = 32,768$	2^{-482}
$2^{17} = 131,072$	2^{-478}
$2^{20} = 1,048,576$	2^{-472}

2.3 Additive Subspaces and Linearized Polynomials

An additive subspace of size 2^k over $\mathbb{F}_2$ is:

$$D = \text{span}_{\mathbb{F}_2} \{\beta_0, \dots, \beta_{k-1}\}$$

A linearized polynomial has the form:

$$L(X) = \sum_{i=0}^k c_i X^{2^i}$$

Frobenius linearity (characteristic 2 only):

$$L(a + b) = L(a) + L(b)$$

because $(a + b)^{2^i} = a^{2^i} + b^{2^i}$.

3 Miko Decomposition

3.1 Vanishing Polynomials over Binary Fields

Theorem 3.1 (Linearized Vanishing Polynomial). *Let $S = \text{span}_{\mathbb{F}_2} \{\beta_0, \dots, \beta_{k-1}\}$ be an additive subspace of $\mathbb{F}_{2^{256}}$ of size 2^k . The vanishing polynomial:*

$$V_S(X) = \prod_{s \in S} (X + s)$$

is a linearized polynomial with exactly $k+1$ non-zero terms. It is constructed via the Artin-Schreier iteration:

$$V_0(X) = X$$

$$V_{i+1}(X) = V_i(X)^2 + \lambda_i V_i(X) \quad \text{where } \lambda_i = V_i(\beta_i)$$

After k iterations, $V_k(X) = V_S(X)$. Evaluation at any point costs $O(k) = O(\log n)$ field operations.

3.2 The Miko Decomposition Algorithm

Definition 3.2 (Miko Decomposition). *Given $R(X) \in \mathbb{F}_{2^{256}}[X]$ of degree $< n = 2^k$ and an $\mathbb{F}_2$ basis $\{\beta_0, \dots, \beta_{k-1}\}$, the Miko decomposition produces a sequence $(Q_0, \dots, Q_{k-1}, r_{\text{final}})$ satisfying the Miko identities:*

$$R_0(X) = R(X)$$

$$R_j(X) = R_{j+1}(X) + V_j(X) \cdot Q_j(X) \quad (j = 0, \dots, k-1)$$

$$R_k = r_{\text{final}} \text{ (constant)}$$

where $V_j(X)$ is the vanishing polynomial of $\text{span}_{\mathbb{F}_2} \{\beta_{j+1}, \dots, \beta_{k-1}\}$.

Theorem 3.3 (Uniqueness). *For any $R(X)$ of degree $< n$ and any $\mathbb{F}_2$ -linearly independent basis, there exists exactly one sequence $(Q_0, \dots, Q_{k-1}, r_{\text{final}})$ satisfying the Miko identities.*

Theorem 3.4 (Exponential Degree Decrease).

$$\deg(Q_j) < \frac{n}{2^{j+1}} = 2^{k-j-1}$$

and

$$\deg(R_j) < \frac{n}{2^j} = 2^{k-j}$$

Table 3: Degree Bounds by Level

Level j	n	$\deg(R_j)$	$\deg(Q_j)$
0	2^k	$< 2^k$	$< 2^{k-1}$
1	2^k	$< 2^{k-1}$	$< 2^{k-2}$
2	2^k	$< 2^{k-2}$	$< 2^{k-3}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$k-1$	2^k	< 2	< 1
k	2^k	< 1	—

3.3 Miko Identity as the Algebraic First Defense

Lemma 3.5 (Miko Identity Check Soundness). *If the prover uses fake quotients $Q'_j \neq Q_j$, the verifier's Miko identity check at s independent challenge points fails with probability at least:*

$$1 - \left(\frac{n}{2^{256}} \right)^s$$

4 Hash Chain Commitment

4.1 Construction

Definition 4.1 (Hash Chain Commitment). *Given quotients $Q_0, \dots, Q_{k-1}$ and remainder r_{final} :*

$$h_k = \text{SHA-256}(\text{"final"} \parallel r_{final_bytes})$$

$$h_j = \text{SHA-256}(h_{j+1} \parallel \text{"level_"} \parallel j \parallel Q_{j_bytes}) \quad (j = k-1, \dots, 0)$$

$$com = h_0 \text{ (32 bytes)}$$

Lemma 4.2 (Hash Chain Binding). *To change any Q_j after publishing com , an adversary must find a SHA-256 preimage. Success probability $\leq 2^{-256}$.*

4.2 The Residual Polynomial Is Intentionally Not Committed

$R(X)$ is excluded from the hash chain. Binding between the committed Q_j and R derives algebraically from the Miko identity check (Lemma 3.6) and Mira cross-binding (Lemma 5.1).

4.3 The Phase 3 Commitment Lock

Lemma 4.3 (Phase 3 Commitment Lock). *Once the prover publishes com in Phase 3 of the proving protocol, no PPT adversary can modify any Q_j without changing com , and consequently the Fiat-Shamir challenges $z_i = \text{SHA-256}(\text{com} \parallel i)$ and Mira parameters derived from com .*

5 Mira: Multi-Instance Recursive Aggregation

5.1 Cross-Binding: Linking Public and Challenge Points

The cross-binding equation prevents the prover from using different Q_j values at public points versus challenge points. For $t = k + 1$ public points $\omega_0, \dots, \omega_{t-1}$:

$$\text{binding} = A_{\text{combined}} + \gamma_1 \cdot C_{\text{combined}} + \gamma_2 \cdot r_{\text{final}}$$

where:

$$A_{\text{combined}} = \sum_{i=0}^{t-1} \beta^i \cdot \sum_{j=0}^{k-1} V_j(\omega_i) \cdot Q_j(\omega_i)$$

$$C_{\text{combined}} = \sum_{j=0}^{k-1} V_j(z) \cdot Q_j(z)$$

$$\gamma_1 = \text{SHA-256}(\text{com} \parallel z \parallel v \parallel 0), \quad \gamma_2 = \text{SHA-256}(\text{com} \parallel z \parallel v \parallel 1)$$

Lemma 5.1 (Mira Cross-Binding Security). *If the prover uses fake $Q'_j \neq Q_j$ that differ from the committed quotients at either public points or the challenge point, the probability that the cross-binding equation holds is $\leq 1/|\mathbb{F}| = 2^{-256}$.*

5.2 Zero-Residual Attack Defense

Attack: Prover claims all $Q'_j = 0$, $r'_{\text{final}} = 0$. Miko identity holds trivially ($0 = 0$). Cross-binding equation becomes $0 = 0$.

Defense — deterministic rejection: The hash chain opening at public points ω_i reveals $Q'_j(\omega_i) = 0$ for all j, i . This forces $A_{\text{combined}} = 0$. The verifier independently computes the expected A-matrix evaluations from the circuit's public inputs w_{pub} :

$$A_{\text{expected}} = \sum_{i=0}^{t-1} \beta^i \cdot (A[i] \cdot w_{\text{pub}})$$

For any real circuit with non-zero public inputs, $A_{\text{expected}} \neq 0$. The verifier checks $A_{\text{combined}} = A_{\text{expected}}$. Since $A_{\text{combined}} = 0 \neq A_{\text{expected}}$, deterministic rejection occurs.

5.3 Mira RLC for RMC Tower Batch Verification

For recursive Miko commitment towers, Mira compresses m internal node identity checks into a single equation per challenge point:

$$Q_{\text{batched}} = \sum_{j=0}^{m-1} \alpha^j \cdot Q_{\text{node}_j}(z)$$

where $\alpha = \text{SHA-256}(\text{com} \mid \text{"rnc_batch"})$.

Lemma 5.2 (Mira RMC Batch Verification Soundness). *If any of the m internal RMC nodes has an incorrect evaluation, the batched check fails with probability at least:*

$$1 - \frac{m}{|\mathbb{F}|}$$

6 RMC Towers: Recursive Miko Commitment

6.1 The Shallow Quotient Problem

For a circuit with $n = 5,000$ constraints ($k = 13$), shallow Miko levels produce quotients with degree up to 2,500. Transmitting all coefficients requires ~ 80 KB.

6.2 Recursive Miko Decomposition

RMC towers recursively apply Miko decomposition until all pieces have degree < 76 (the base polynomial threshold). The resulting binary tree encodes the Miko decomposition structure.

Miko identity at every internal node:

$$Q_{\text{parent}}(z) = Q_{\text{left}}(z) + V(z) \cdot Q_{\text{right}}(z)$$

6.3 Dual Binding Theorem

Theorem 6.1 (RMC Tower Dual Binding). *Let $B = (B_1, \dots, B_m)$ be the base polynomials of an RMC tower with root commitment c , each of degree $< d$. An adversary providing fake base polynomials $B' \neq B$ such that:*

1. $\text{com}(B') = \text{com}(B)$ (same root commitment)
2. The Miko identity holds at s independent random challenge points

succeeds with probability at most:

$$\min \left(2^{-256}, \left(\frac{d}{2^{256}} \right)^s \right)$$

6.4 Corollary: Hash-Independent Security

Corollary 6.2. *Even if SHA-256 is completely broken (instant collisions found), the algebraic layer (Miko identity check) remains secure. This distinguishes RMC towers from Merkle trees, whose security collapses entirely if the hash function breaks.*

7 Complete Security Analysis

7.1 The Prover's Only Degree of Freedom

The prover selects a witness w . From w , everything follows deterministically:

1. $R(X) = \sum_{i=0}^{n-1} r_i X^i$ where $r_i = (A[i] \cdot w) \otimes (B[i] \cdot w) \oplus (C[i] \cdot w)$
2. Miko decomposition: $R \rightarrow (Q_0, \dots, Q_{k-1}, r_{\text{final}})$ unique by Theorem 3.3
3. Hash chain: $\text{com} = \text{SHA-256-chain}(Q_j, r_{\text{final}})$
4. Fiat-Shamir challenges: $z_i = \text{SHA-256}(\text{com} \parallel i)$, $\beta, \gamma_1, \gamma_2, \delta = \text{SHA-256}(\text{com} \parallel \dots)$
5. Evaluations: $Q_j(z_i), R(z_i)$ — deterministic from the committed Q_j

7.2 Three Exhaustive Attack Strategies

Theorem 7.1 (Complete Soundness). *Let C be an unsatisfiable circuit with $n \leq 2^{17}$ constraints over $\mathbb{F}_{2^{256}}$. For any PPT adversary A making q random oracle queries:*

$$\Pr[\text{Verifier}^H(C, A^H(C)) = \text{ACCEPT}] \leq \frac{t}{2^{256}} + \frac{m}{2^{256}} + O\left(\frac{q^2}{2^{256}}\right)$$

where $t = k + 1$ and m is the number of Mira-batched RMC nodes. For practical parameters, this yields $\sim 2^{-250}$ security.

Note: The dominant terms are the Mira folding errors $t/2^{256}$ and $m/2^{256}$, which for $n = 2^{17}$ give $\sim 2^{-250}$. This still exceeds the NIST 128-bit requirement by 122 bits.

7.3 The Three Attack Strategies

7.3.1 Strategy 1: Use Real $R \neq 0$ from Invalid Witness

The prover selects w such that $R \neq 0$. The verifier checks $R(z_0) = R(z_1) = 0$. By Schwartz-Zippel:

$$\Pr[R(z_0) = 0 \wedge R(z_1) = 0] \leq \left(\frac{n}{2^{256}}\right)^2$$

7.3.2 Strategy 2: Use Fake $Q'_j \neq Q_j$

The prover substitutes fake quotients in the hash chain. The verifier's Miko identity check catches this with probability $\geq 1 - (n/2^{256})^2$. Even if this check is bypassed, the Mira cross-binding catches the forgery with probability $\geq 1 - 2^{-256}$.

7.3.3 Strategy 3: Zero Residual Attack

The prover sets all $Q'_j = 0$, $r'_{\text{final}} = 0$. The Miko identity holds trivially. The public-point consistency check catches this deterministically: $A_{\text{combined}} = 0 \neq A_{\text{expected}}$.

7.4 Witness Grinding Bound

The prover can try T different witnesses. Each trial succeeds with probability $\leq (n/2^{256})^2$. By the union bound:

$$\Pr[\text{any success}] \leq T \cdot \left(\frac{n}{2^{256}}\right)^2$$

T is bounded by the number of ROM queries q , contributing the $O(q^2/2^{256})$ term.

8 First Application: PMM with Recursive Composition

The three primitives compose into PMM, demonstrating native recursion: PMM proves PMM verification over the same field with no additional cryptography.

Table 4: Recursive Proof Architecture

Inner Proof π_1	Outer Proof π_2	On-Chain Verifier
Full verification of original circuit	Verifier circuit ($\sim 5,000$ constraints)	One
Variable size (~ 400 -500 bytes)	Constant ~ 140 bytes	~ 140 bytes
Off-chain	On-chain	Post-quantum

8.1 Proof Size Breakdown

For the outer proof π_2 (verifier circuit, $k = 13$, all quotients zero when π_1 is valid):

Table 5: Outer Proof Size Breakdown

Component	Size
<code>com₂</code>	32 bytes
<code>binding₂</code>	32 bytes
<code>all_quotients_folded</code> (Mira over all 13 levels)	32 bytes
<code>remainder₂</code>	32 bytes
Counts/overhead	~ 12 bytes

Total: ~ 140 bytes — constant, independent of original circuit size.

8.2 Recursive Soundness Preservation

Theorem 8.1 (Recursive Soundness Preservation). *If the primitives are sound for the verifier circuit, then PMM with recursive composition is sound for any circuit.*

Each recursion layer adds independent Schwartz-Zippel checks. After ℓ layers, soundness error:

$$\epsilon_\ell \leq \left(\frac{n}{2^{256}}\right)^{2^{\ell+1}}$$

Proof. The outer proof π_2 does not trust π_1 — it verifies it. The outer circuit is the PMM verifier, which checks π_1 's hash chain, Miko identity, and Mira cross-binding. If π_1 is invalid, the verifier

rejects, the outer circuit's residual $R_2 \neq 0$, and π_2 cannot be generated (by Theorem 7.1 applied to the outer circuit).

Each recursion layer adds $s = 2$ independent challenge points with fresh Fiat-Shamir randomness. The challenge sets for different layers are independently distributed in the Random Oracle Model, using distinct domain separators. Therefore, the soundness errors multiply across layers. $\square$

9 Post-Quantum Security

Theorem 9.1 (Post-Quantum Security). *All three primitives are secure against quantum adversaries. Shor's algorithm provides no advantage. Grover's algorithm provides at most a quadratic speedup on SHA-256 preimage search.*

Proof. Exhaustive component analysis:

- **Field arithmetic:** XOR, carryless multiply over $\mathbb{F}_{2^{256}}$ — No DLOG in characteristic 2.
- **Miko decomposition:** Polynomial division — Not exponentiation. No hidden subgroup problem.
- **Hash chain:** SHA-256 compression function — Grover only: $2^{256} \rightarrow 2^{128}$ preimage search.
- **Schwartz-Zippel:** Information-theoretic. No quantum algorithm can increase the number of roots of a polynomial over a finite field.
- **Fiat-Shamir:** SHA-256(com | i) — QROM-secure via Unruh (2015).
- **Mira RLC:** Linear combination — No quantum speedup for addition or scalar multiplication.
- **RMC towers:** SHA-256 chain + Miko identity — Algebraic binding (Corollary 6.2) is information-theoretic.

Table 6: Quantum Attack Surface Analysis

Component	Operation	Quantum Vulnerability
Field arithmetic	XOR, carryless multiply over $\mathbb{F}_{2^{256}}$	None — No DLOG in char 2
Miko decomposition	Polynomial division	None — Division, not exponentiation
Hash chain	SHA-256(h_{j+1} Q_j)	Grover only: $2^{256} \rightarrow 2^{128}$
Schwartz-Zippel	$\Pr[f(z) = 0] \leq d/2^{256}$	None — Information-theoretic
Fiat-Shamir	$z_i = \text{SHA-256}(\text{com} \mid i)$	QROM-secure via Unruh (2015)
Mira RLC	$\sum \alpha^j \cdot v_j$	None — Linear combination
RMC towers	SHA-256 + Miko identity	None beyond hash chain

10 Performance Characteristics

10.1 Prover Complexity

Table 7: Prover Complexity

Residual Computation	$O(n)$
Miko Decomposition	$O(n \log^2 n)$
RMC Tower Construction	$O(n \log n)$
Hash Chain Commitment	$O(n \log n)$
Evaluation at Challenge Points	$O(s \cdot n)$
Total	$O(n \cdot (s + \log n)) \approx O(n \log n)$

10.2 Verifier Complexity

Table 8: Verifier Complexity

Proof Parsing	$O(k)$
Hash Chain Reconstruction	$O(n)$
Challenge Derivation	$O(s)$
Miko Identity Verification	$O(s \cdot (k^2 + n))$
RMC Tower Traversal	$O(k \log k)$
Total	$O(s \cdot n + s \cdot k^2)$

10.3 Proof Size

Table 9: Proof Size

Inner Proof π_1	$\sim 400\text{--}500$ bytes (variable)
Outer Proof π_2	~ 140 bytes (constant)

11 Conclusion

We have introduced three post-quantum transparent primitives over binary fields: Miko Decomposition (deterministic low-degree testing via a 2-challenge Miko identity), Hash Chain Commitment (Merkle-free binding with algebraic residual binding and a Phase-3 lock for native recursion), and Mira (RLC-based aggregation for cross-binding, batch tower verification, and constant-size proofs). The dual-layer security — algebraic Schwartz-Zippel plus SHA-256 — guarantees hash-independent soundness. When composed into PMM, the primitives deliver ~ 140 -byte on-chain proofs with native recursion, no trusted setup, and post-quantum security in the random oracle model.

References

1. Schwartz, J. T. & Zippel, R. Probabilistic polynomial testing and applications to matrix identities. *Journal of the ACM*, 27(4), 701-717, 1980.

2. Fiat, A. & Shamir, A. How to prove yourself: Practical solutions to identification and signature problems. *CRYPTO 1987*.
3. Unruh, D. Revocable quantum timed-release encryption. *Journal of Cryptology*, 29(4), 955-1009, 2015.
4. Ben-Sasson, E., Bentov, I., Horesh, Y., & Riabzev, M. Scalable, transparent, and post-quantum secure computational integrity. *EUROCRYPT 2018*.
5. Groth, J. On the size of pairing-based non-interactive arguments. *EUROCRYPT 2016*.
6. Gabizon, A., Williamson, Z. J., & Ciobotaru, O. PLONK: Permutations over Lagrange-bases for Oecumenical Noninteractive arguments of Knowledge. *Cryptology ePrint Archive*, Report 2019/953.
7. Lidl, R. & Niederreiter, H. *Finite Fields* (2nd ed.). Cambridge University Press, 1997.
8. NIST SP 800-38D. Recommendation for Block Cipher Modes of Operation: GCM and GMAC.